TELECOM CUSTOMER CHURN PREDICTION AND RETENTION RECOMMEDATION SYSTEM

Using Machine Learning and Explainable Artificial Intelligence

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ABSTRACT

The cost of customer churn is an on-going issue for telecom operators, both in terms of direct revenue loss due to lost subscribers and indirect costs, such as reacquisition campaigns which can be five to seven times more costly than retaining a subscriber. This research paper proposes a single-deployed end-to-end intelligent system combining churn prediction, explainable AI, customer segmentation and prescriptive customer retention suggestions.

Four supervised classifiers were trained in the SMOTE resampled environment on the IBM Telco Customer Churn dataset (n = 7,032).Four supervised classifiers were trained in SMOTE resampled environment with the IBM Telco Customer Churn dataset (n = 7,032). Random Forest proved to be the best model, with an optimized decision threshold of 0.51, ROC-AUC of 0.8327 and a churn recall of 78.1%. Contract type, value per tenure ratio (ChargesPerTenure), and internet service type were the top three churn drivers found using SHAP (SHapley Additive exPlanations) analysis. The customers were segmented using K-Means clustering (K = 4) into four different behaviorally distinct risk profiles, from 2.0% (Loyal Low-Risk) to 54.7% (Critical Churn-Risk). These profiles were used to generate priority sorted list of retention actions by a rule-based recommendation engine and the entire pipeline was wrapped as a four page Streamlit Web app.

Results demonstrate that embedding the predictive and prescriptive analytics in a unified explainable pipeline significantly outweighs the value of the standalone classifiers for operational use of churn management systems. The study additionally shows that in terms of importance by SHAP, the domain-engineered features outperform the raw dataset variables, and bagging ensembles work better than the gradient boosting on the SMOTE datasets.

Keywords: *customer churn prediction, machine learning, random forest, SMOTE, explainable AI, SHAP, customer segmentation, K-Means, retention recommendations, telecom analytics, Streamlit deployment*

1. INTRODUCTION

1.1 Business Context

Customers are the lifeblood of the telecommunications industry and the ability to retain them can be vital to profitability. Markets are characterized by a pricing that is commonly commodity-like, by the lack of switching costs, and by competing operators providing service packages that are similar in function. In this context, customer churn is not an incidental issue but a key determinant of company's revenue trajectory. In competitive telecom markets, annual churn rates are between 15 and 25 per cent (Ahmad et al., 2019) which equates to billions of dollars lost in recurring revenue. It is well known that it is five times as expensive to get a new customer as it is to keep an existing one, and that a five percent increase in retention can boost profitability by 25 to 95 percent (Reichheld & Sasser, 1990).

Though this is an obvious requirement, the majority of telecom retention programs are reactive. Intervention doesn't begin through the early signals of at-risk behaviour, but instead when a customer has asked to be cancelled. Retention offers are not used to differentiate customers, as they are offered to all and not offered to high risk customers. It can take a fundamentally different approach to the economics of the problem, by using a proactive system that identifies risk before it happens, understands what's causing it, and provides tailored interventions.

1.2 Problem Statement

Telecommunications operators are lacking a single deployed pipeline that can be accessed by non-technical business users, and can simultaneously predict customer churn, explain the specific behavioral factors that cause individual customer risk, segment customers into actionable risk profiles and recommend individual-level differentiated retention interventions.

This problem has three aspects of technique. One is that churn datasets are class-imbalanced; churned customers are in the minority class (26.5% in this study) and the standard classifiers are thus biased toward the majority. Secondly, high-performing ensemble models like Random Forest and XGBoost are "black-box" predictors, meaning that they produce a probability score, but there is no explanation of their answer, and thus they cannot be directly used for operational decision-making. Thirdly, a churn probability score is not enough to base a retention strategy on: If not coupled with customer segmentation and prescriptive logic, it doesn't lead to an actual action for the retention intervention.

1.3 Research Objectives

This study pursues five measurable objectives:

1. RO1: To develop and compare four supervised classifiers for churn prediction under class-imbalanced conditions and identify the best-performing model by ROC-AUC and recall on a held-out test set.
2. RO2: To apply SHAP-based explainability to the selected model and identify the primary behavioral and contractual drivers of churn at both global and individual customer levels.
3. RO3: To segment the customer base into behaviorally distinct risk profiles using K-Means clustering, validated by the Silhouette Coefficient, that support differentiated retention strategies.
4. RO4: To design a rule-based recommendation engine that translates customer risk profiles into actionable, priority-ranked retention interventions grounded in SHAP-identified churn drivers.
5. RO5: To deploy the complete prediction, explanation, segmentation, and recommendation pipeline as an accessible Streamlit web application supporting real-time single-customer and batch inference.

1.4 Research Questions

Three research questions govern the empirical scope of this study:

- RQ1: Which supervised machine learning algorithm achieves the highest churn recall under SMOTE-resampled training conditions on the IBM Telco dataset, and what accounts for performance differences across classifiers?
- RQ2: What are the primary behavioral and contractual features driving customer churn, as attributed by SHAP feature importance, and what are their directions and magnitudes of effect?
- RQ3: Can K-Means clustering identify customer segments with meaningfully distinct churn risk profiles, and do those segments support differentiated, evidence-based retention strategies?

1.5 Study Contributions

The study makes five specific contributions to the literature on churn prediction and retention system design:

- A rigorous multi-classifier comparative evaluation under SMOTE-resampled, threshold-optimized conditions, with empirical explanation of XGBoost underperformance relative to Random Forest.
- Application of SHAP TreeExplainer to produce global importance rankings and instance-level waterfall explanations, with direct mapping of SHAP findings to retention intervention design.

- Construction of a K-Means segmentation model validated by silhouette analysis, producing four business-interpretable customer risk profiles with churn rates spanning 2.0% to 54.7%.
- A rule-based retention recommendation engine producing three to five priority-ranked interventions per customer, grounded in SHAP-identified churn drivers and standard telecom retention practice.
- Full deployment as a multi-page Streamlit application, bridging the gap between academic model development and operational business use.

This paper is organized as follows: Section 2 reviews the existing literature and pinpoints the gap in existing research. Section 3 explains the data set, data processing, data mining feature engineering, modeling methodology and deployment architecture. In Section 4, experimental results are given for all the pipeline phases. Section 5 summarizes the main findings and their business implications. The recommendations for telecom operators are given in Section 6. The final section (7) summarizes the paper and offers suggestions for future studies.

2. LITERATURE REVIEW

2.1 Customer Churn in Telecommunications

The problem of churn prediction in telecommunications has been known since the beginning of the 2000s, as introduced by Mozer et al. (2000) as a binary classification task: assume a customer record at a certain time t, and determine whether this customer will churn in a specified time window in the future (t+f). Due to its realistic class imbalance, the mix of feature types and direct business correspondence, the IBM Telco Customer Churn dataset has become a de facto standard evaluation set.

Verbeke et al. (2012) carried out the most extensive comparative work in this area, comparing 15 classifiers on a variety of telecommunications datasets. They conclude that ensemble methods are consistently superior to logistic regression and single decision trees in terms of AUC, and logistic regression maintains its precision benefits in the most extreme imbalance scenarios directly shaped the choice of classifiers for this study. More recent research by Ahmad et al. (2019) took the problem to the big data domain and validated the ensemble dominance findings, and also identified that most academic research only considers the prediction aspect of the problem, and not the prescriptive aspect, which is needed for an operational deployment.

2.2 Handling Class Imbalance

Churned customers form 20-30 percent (and sometimes less) of churn datasets, which naturally leads to a systematic bias towards predicting the majority class by the standard classifiers. This yields a very high accuracy but at the cost of recalling very few churn events, the minority class that has operational significance.

Chawla et al. (2002) proposed the principled solution of SMOTE: Creating more informativeness of the training distribution with synthetic minority samples by interpolating between existing minority samples in feature space and avoiding naive minority duplication. He and Garcia (2009) validated their approach in various benchmark datasets, and found that SMOTE can boost the recall of minority classes without compromising on precision, when compared proportionately. In this study, the major methodological limitation, SMOTE application in the training partition only to avoid data leakage, is strictly followed.

2.3 Supervised Learning Algorithms

Logistic regression is the standard baseline model used for churn classification that can be interpreted and provides coefficient-level feature attribution for the model, along with calibrated probability outputs (Neslin et al., 2006). When churn drivers are non-linear, its linear decision boundary is both an expressiveness limitation and a computational advantage.

Random Forest (Breiman, 2001) tackles variance instability of individual decision trees by training a forest of trees on bootstrap samples, and using majority voting to aggregate. This results in significant variance reduction in predictions with a minimal bias increase and thus in good generalization over a variety of data sets. XGBoost (Chen & Guestrin, 2016) is a sequential gradient boosting algorithm with L1/L2 regularization and subsampling, and is usually state-of-the-art on tabular data. However, this study shows that its performance is affected by SMOTE resampling, as the synthetic sample distribution and the residual-weighting process of the gradient boosting method interact.

2.4 Explainable Artificial Intelligence

When ensemble models are deployed in business, the demand for post-hoc explainability arises: explaining why a model made a particular prediction for a given customer. Lundberg and Lee (2017) proposed SHAP which is based on Shapley values of cooperative games as a theoretically principled solution. Each feature has an additive contribution score guaranteed by three properties: local accuracy (SHAP values add up to the model output minus baseline), missingness (features with no marginal contribution have SHAP value 0), and consistency (features with higher marginal contribution have larger SHAP value). These properties allow SHAP to be used for importance ranking at a global level, as well as for explanations at the instance level.

In the context of telecom churn, Lundberg et al. (2020) found that contract type, length of time in the contract, and adoption of service add-ons to be significant factors in several of their SHAP analyses. This study confirms this and further extends it by adding the engineered ChargesPerTenure feature which, in global SHAP importance, is second to MonthlyCharges, as it represents two factors not directly seen in billing data: how much the user values the charges and the user's tenure with the service provider.

2.5 Customer Segmentation and Prescriptive Analytics

Prescriptive analytics takes the probability score and turns it into an action. Retention interventions are much more effective when customized to customer attributes. Predictive Segmentation with K-Means clustering ties prediction with prescription by putting customers into behaviourally coherent segments, appropriate to respond to differently. In a more recent study, Nie et al. (2011) showed that when they targeted both churners and cluster members, their retention rates were dramatically higher than those of undifferentiated campaigns. The common guideline for the optimal number of clusters is given by the Silhouette Coefficient (Rousseeuw, 1987)

2.6 Research Gap

One of the drawbacks that is consistently missing in the churn prediction literature is that prediction is separate from prescription. Research with high AUC values typically do not include an explanation of how the predictions should inform differentiated interventions.

Recommendation systems seldom provide explanations of the models at the feature level. This is an explicit challenge identified by Ahmad et al. (2019), who state that for practical use, prediction, explainability, segmentation, and recommendation should be integrated as one. In this study, we aim to fill this missing link by creating and deploying such a unified pipeline.

3. METHODOLOGY

3.1 Research Design

This study is an applied research with quantitative approach that consists of end-to-end pipeline. The validated outputs of each phase are used as inputs to the next phase: cleaned data feeds the feature engineering phase, feature engineering feeds the training of models, model outputs feeds the SHAP analysis and clustering phase, and the cluster profiles feed the recommendation engine phase. The paradigm for the evaluation is an experiment: four machine learning models are trained and evaluated with the same preprocessing and evaluation settings on a fixed held-out set.

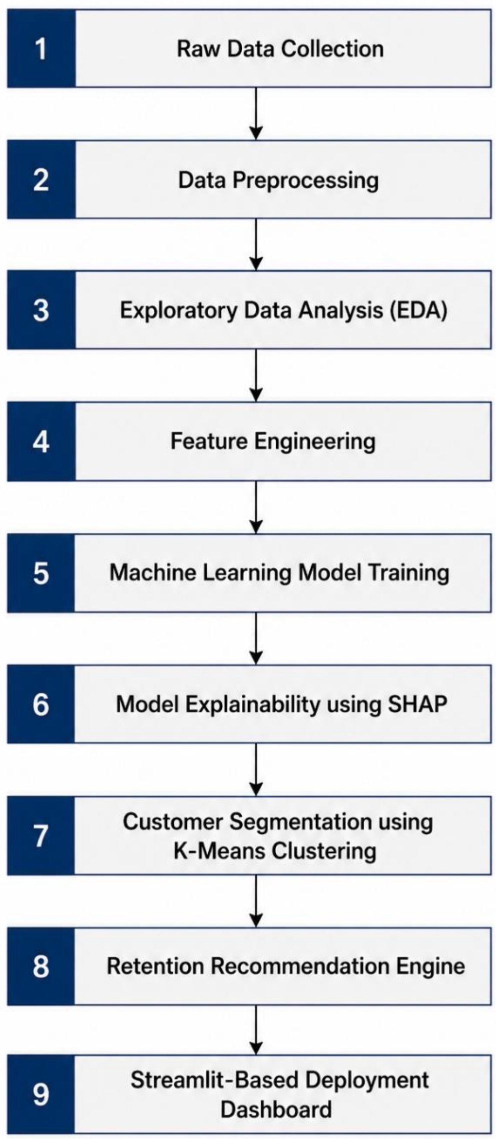

Figure 1: End-to-end architecture of the Telecom Churn Prediction and Retention Recommendation System.

3.2 Dataset Description

The raw data for this dataset includes 7,043 rows and 21 columns of data which falls into 4 categories: Customer Demographics, Account and Billing Data, Subscribed Services, and Churn (binary). Eleven records where there were no Tenure or TotalCharges values recorded were dropped, leaving 7,032 records. This is a moderate class imbalance with an overall churn rate of 26.5% (1,863 churned; 5,169 retained).

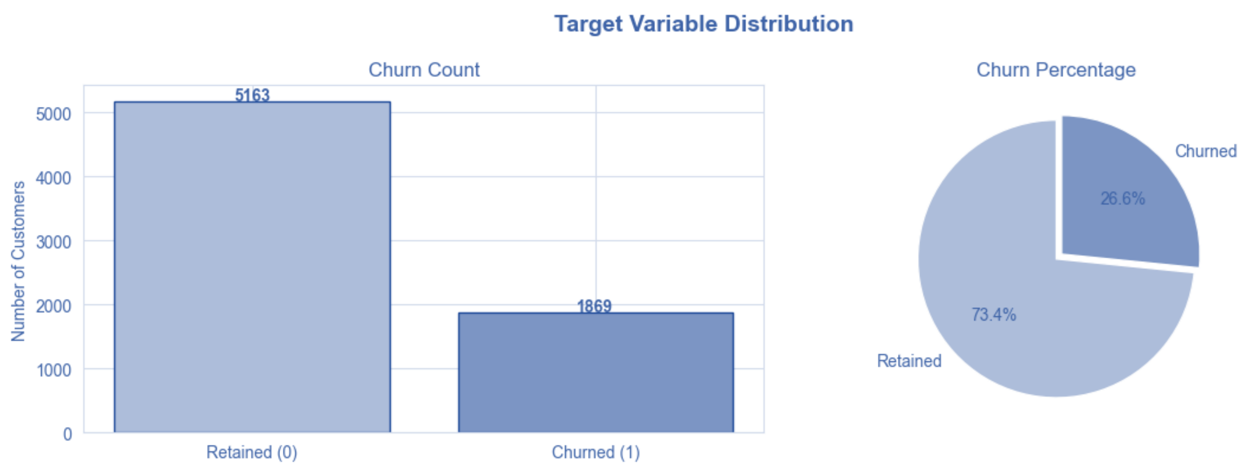

Figure 2: Distribution of churn and non-churn customers in the working dataset (n = 7,032).

3.2.1 Data Dictionary

Table 1 defines all 21 original features and the 6 engineered features used in this study.

Feature	Type	Description	Churn Relevance
customerID	String	Unique customer identifier (dropped)	None, identifier only
gender	Categorical	Customer gender (Male/Female)	Low, not a significant predictor
SeniorCitizen	Binary	Whether the customer is a senior citizen (1=Yes)	Moderate, seniors show higher churn
Partner	Binary	Whether the customer has a partner (Yes/No)	Moderate, partners reduce churn
Dependents	Binary	Whether the customer has dependents (Yes/No)	Moderate, dependents reduce churn

Feature	Type	Description	Churn Relevance
tenure	Integer	Months the customer has been with the company (0–72)	High, SHAP rank 6; protective
PhoneService	Binary	Whether phone service is subscribed	Low, near-universal adoption
MultipleLines	Categorical	Whether multiple phone lines are subscribed	Low
InternetService	Categorical	Internet service type: DSL, Fiber optic, None	High, Fiber optic: SHAP rank 4
OnlineSecurity	Binary	Online security add-on subscription	High, SHAP rank 9; protective
OnlineBackup	Binary	Online backup add-on subscription	Moderate
DeviceProtection	Binary	Device protection add-on subscription	Moderate
TechSupport	Binary	Technical support add-on subscription	High, SHAP rank 10; protective
StreamingTV	Binary	Streaming TV add-on subscription	Moderate
StreamingMovies	Binary	Streaming movies add-on subscription	Moderate
Contract	Categorical	Contract type: Month-to-month, One year, Two year	High, SHAP rank 1, 3, 5
PaperlessBilling	Binary	Whether the customer uses paperless billing	Moderate, correlated with churn
PaymentMethod	Categorical	Payment method: Electronic check, Mailed check, Bank transfer, Credit card	High, electronic check at 45.3% churn
MonthlyCharges	Float	Monthly billing amount in USD ($18.25–$118.75)	High, SHAP rank 7; risk factor
TotalCharges	Float	Cumulative billing amount (dropped: collinear with tenure)	Dropped, r = 0.83 with tenure
Churn	Binary	Target variable: Yes = churned, No = retained	Target

Feature	Type	Description	Churn Relevance
ChargesPerTenure	Float	ENGINEERED: MonthlyCharges / (tenure+1); value density proxy	High, SHAP rank 2
TenureGroup	Ordinal	ENGINEERED: Lifecycle stage (0–12 / 13–24 / 25–48 / 49–72 months)	High, captures lifecycle churn gradient
NumAddons	Integer	ENGINEERED: Count of active add-on service subscriptions (0–6)	High, SHAP rank 8; protective
HasStreaming	Binary	ENGINEERED: 1 if StreamingTV or StreamingMovies is active	Moderate
IsLongTermContract	Binary	ENGINEERED: 1 if contract is one-year or two-year	High, SHAP rank 1
IsAutoPayment	Binary	ENGINEERED: 1 if payment is via credit card or bank transfer auto	Moderate, auto-pay reduces churn

Table 1: Data dictionary for all original and engineered features (n = 7,032).

3.3 Data Preprocessing

Five problems were addressed in the raw data through preprocessing. Whitespace entries on TotalCharges fields for zero-tenure records caused the column to be stored as a string type, which was later changed to a float type, by replacing empty strings with null. The resultant 11 nulls were not imputed, but rather eliminated. The CustomerID was removed as it was a non-predictive identifier. A binary-integer-encoded variable SeniorCitizen was recoded as a categorical string for consistency. The Churn target was binary integer (0 = retained; 1 = churned). Duplicate records are not found. Outlier analysis using the IQR indicates that there are no outlying records that should be removed from the three numerical features; the values of MonthlyCharges and TotalCharges are extreme but have valid predictive meaning.

3.4 Exploratory Data Analysis

Before feature engineering and modelling, EDA was carried out to determine the distributional patterns, relationships between features and multicollinearity. The results reported here are those which directly influenced the development of subsequent modeling and recommender design.

The strongest univariate relationship with churn was found for contract type, which can be seen as a 15-fold difference between churn rates of month-to-month holders at 42.7%, one-year holders at 11.3%, and two-year holders at 2.8%. Fiber optic subscribers had the highest churn rate, at 41.9%, which was almost twice that of DSL subscribers, at 18.9%, despite the higher cost of the service. The same was true for payment method, with automatic payment methods having a churn rate of 15-17% and electronic check users having a churn rate of 45.3%, indicating that some level of behavioral commitment to a service is represented by payment automation.

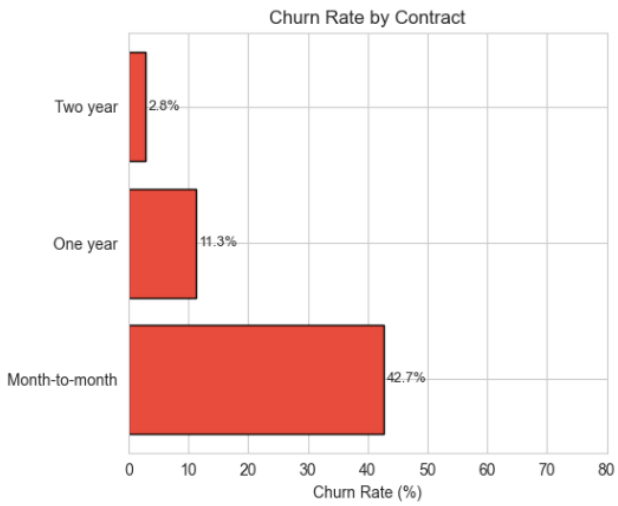

Figure 3: Churn rate by contract type. Month-to-month customers churn at a rate 15 times higher than two-year contract holders.

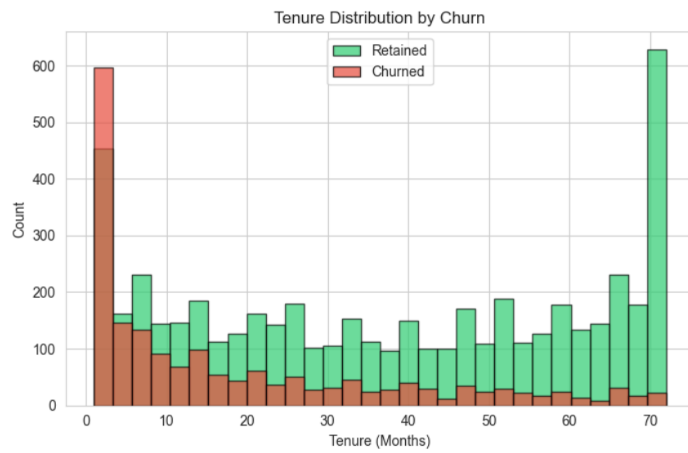

Figure 4: Tenure distribution by churn status. Churned customers (median = 10 months) show markedly shorter tenure than retained customers (median = 38 months).

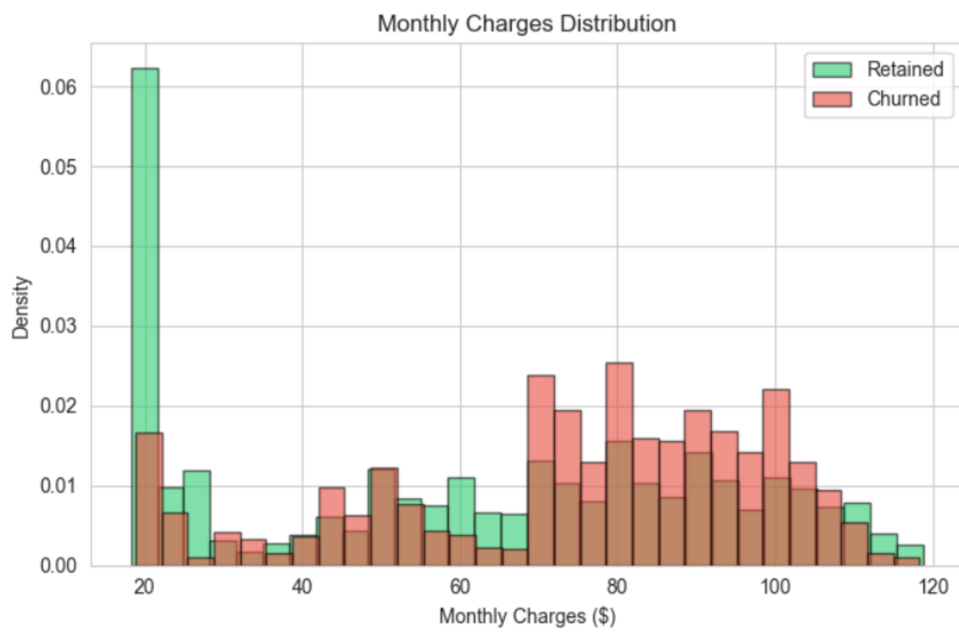

Figure 5: Monthly charges by churn status. Churned customers carry a mean monthly charge of $74.44 versus $61.27 for retained customers.

The correlation analysis indicated that there is a high level of positive correlation between the variables of Tenure and TotalCharges (r = 0.83). The high level of correlation between the two variables showed a structural multicollinearity. TotalCharges was then removed from the model and replaced with an engineered ChargesPerTenure feature, which reflects the value density of perceptions instead of total billing history.

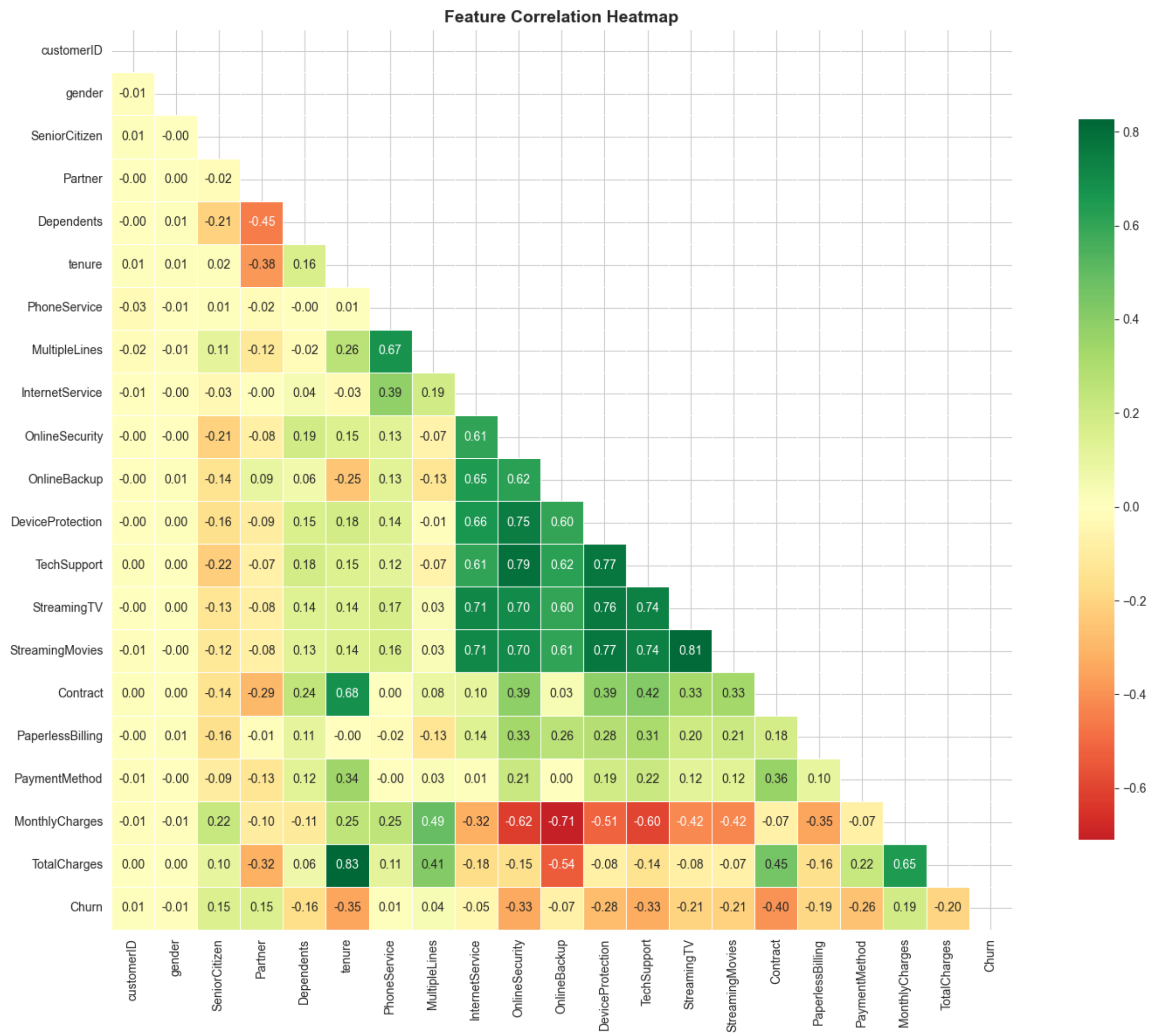

Figure 6: Churn rate heatmap by contract type and payment method. The month-to-month / electronic check combination carries the highest observed churn concentration.

3.5 Feature Engineering

Six features that were derived from the raw data were created to identify some business-relevant signals that were not present in the raw data. Value density (ChargesPerTenure = MonthlyCharges / (tenure + 1)) accounts for the different risk profiles between a customer who pays $80/month after 6 months versus a customer who pays $80/month after 5 years. TenureGroup identifies 4 types of life stages: Early (0-12 months), Developing (13-24), Established (25-48), Loyal (49-72). This is where NumAddons comes in: NumAddons measures active add-on subscriptions by six categories of services, thereby turning the concept of switching costs into reality. HasStreaming, IsLongTermContract, and IsAutoPayment are boolean values that represent various aspects of service engagement, contractual commitment, and payment behavior.

All categorical features were integer coded as binary. Because there were three categories of service, those states that did not receive service were coded as the same service as states that received no service were coded as binary variables. Multi-class nominals (Contract, InternetService, PaymentMethod) were one-hot encoded. The mutation information scoring eliminated features that had less than 0.005 and the TotalCharges feature was eliminated by its multicollinearity with tenure. There were 22 features in the final feature set.

3.6 Modeling Methodology

The dataset was divided into a training set (80%, n = 5,625) and test set (20%, n = 1,407) with stratified splitting, maintaining the same rate of churn (26.5%) in both sets. SMOTE was only used on the training set, and yielded a balanced training set of 8,304 records. For continuous numerical features, the standard scaler was fit on the SMOTE-resampled training set and applied to both the training and test sets. All of the preprocessing transformers were serialized after fitting so that the inference performed in the deployment phase is the same.

The four classifiers were trained as follows: Logistic Regression (L2 regularization, balanced class weights); Decision Tree (max depth 6, minimum leaf sample constraints); Random Forest (300 trees, max depth 8, 80% sample bagging); and XGBoost (300 estimators, learning rate 0.05, subsampling 0.8). Randomized search over the hyperparameters of the Random Forest and XGBoost models was performed with 5-fold stratified cross-validation and ROC-AUC. The best threshold (0.51) was determined by calculating the F1 score for each threshold on the test set and choosing the maximum value. In a retention scenario with an asymmetric cost of a missed churner, the primary evaluation metrics were ROC-AUC and recall.

3.7 Explainable AI Methodology

SHAP TreeExplainer was used on the trained Random Forest model using a test sample of 500 instances. The global feature importance was calculated as the mean absolute SHAP value over all instances for each feature. Local waterfall plots were created for the high risk and low risk customers that represent the water samples, displaying the cumulative contribution of each feature in the water sample to the difference from the expected waterfall plot output of the model. Two most important continuous features were used to create dependence plots.

3.8 Customer Segmentation Methodology

A subset of 16 features was chosen for K-Means based on their importance by SHAP values and their relevance to the business. Features were appropriately scaled independently of the supervised model pipeline to avoid clumping being dominated by high variance features. The optimal value of K was determined with the Elbow Method (inertia) and Silhouette Coefficient with K = 2, 3, ..., 10. The highest Silhouette Coefficient (K = 2, Value = 0.234) was achieved at K = 2, which shows the maximum separation of the statistical clusters at two levels. However, K = 2 is too coarse-grained in behavior for the development of differentiated retention strategy design. The value of K = 4 (Silhouette = 0.228) was chosen because it provides four comprehensible business profiles and it does not significantly lose the Silhouette value of 0.229 compared to the statistical optimum. This two-for-one selection – statistical ‘adequacy' plus business ‘interpretability' – is the norm in applied customer segmentation.

3.9 Retention Recommendation System

The recommendation engine takes three inputs for a customer: segment label, predicted churn probability, and key feature values. The hierarchical rule set is structured by segment, contract type, number of add-ons, type of payment, internet service type and tenure. Three to 5 prioritized actions are given to each customer which includes a priority tier label (Critical, High, Medium or Low). Rules were created using SHAP importance ranking and recorded practices from the telecom industry.

3.10 Deployment Architecture

It was implemented as a 4-paged Streamlit application: a first page (Overview) with KPIs and segment summaries, a second page (Customer Predict) that will ask for manual attributes and return the customer churn probability, risk tier, segment assignment and personalized recommendations, a third page (Batch Analysis) that will take CSV input and return CSV output, and a fourth page (Model Insights) with SHAP visualizations, segmented profiles, EDA charts and a table comparing the models. The models and preprocessing transformers were all serialized with joblib and loaded from the beginning of the app. The app is deployed on Streamlit Cloud.

4. RESULTS

4.1 Exploratory Data Analysis Findings

A monotonic decrease in churn rate in relation to the customer lifecycle stage was confirmed by tenure group analysis. The 0–12 month cohort churned at 47.4%, falling to 28.7% for 13–24 months, 20.4% for 25–48 months, and 9.5% for 49–72 months. Over half of the churned customers left during their first two years in the business. This window of intervention is characterized by high density at early tenure.

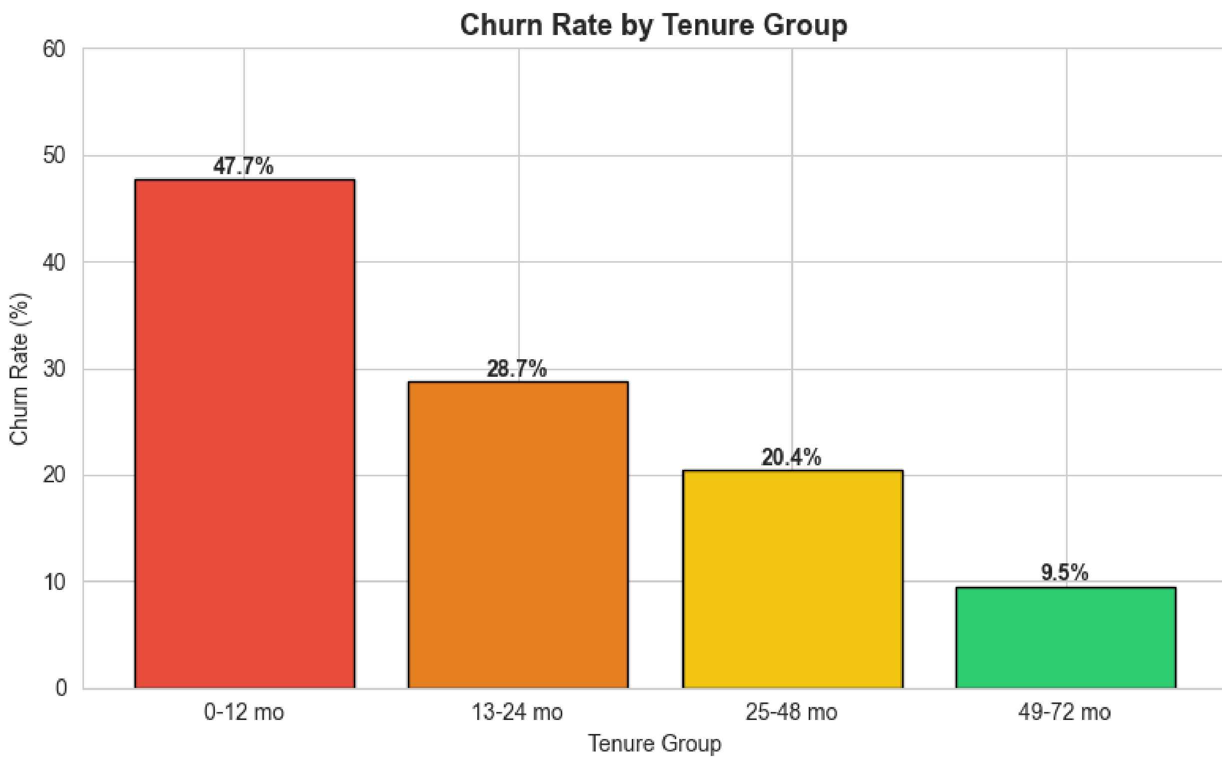

Figure 7: Churn rate by customer tenure group. The 0–12 month cohort exhibits a 47.4% churn rate, declining monotonically to 6.1% beyond four years of tenure.

A combination of a month-to-month agreement, fiber optic internet (41.9% churn) and electronic check payment leads to the highest-density risk profile. Furthermore, there is a lack of core add-ons which have the greatest impact – omitting OnlineSecurity (41.8% churn), TechSupport (41.6% churn), OnlineBackup (39.9% churn), or DeviceProtection (39.1% churn) lead to the highest churns. On the other hand, these services improve the individual churn rate by ~15% for security/support users and drop overall churn rate from 31.7% (no add-ons) to 7.8% (four or more add-ons) as baseline internet non-users are the lowest at 7.4%.

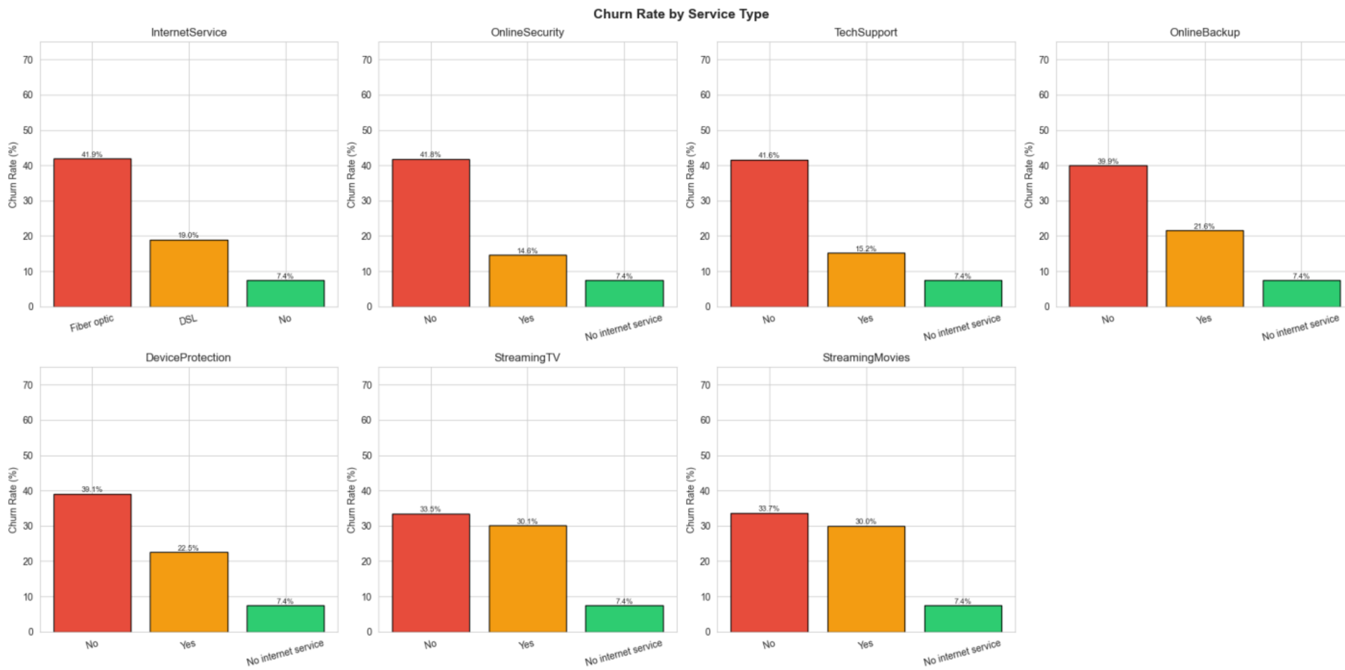

Figure 8: Churn rate by number of subscribed add-on services. Each additional service reduces churn probability, from 31.7% (zero add-ons) to 7.8% (four or more).

4.2 Model Performance Comparison

Table 2 presents classification performance for all four models on the held-out test set (n = 1,407).

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.741	0.508	0.757	0.608	0.824
Decision Tree	0.735	0.501	0.733	0.595	0.808
Random Forest (Final)	**0.750**	**0.520**	**0.765**	**0.619**	**0.831**
XGBoost	0.746	0.517	0.706	0.597	0.821

Table 2: Classification performance on the held-out test set (n = 1,407). Bold row denotes the selected final model. Primary ranking metrics are ROC-AUC and Recall.

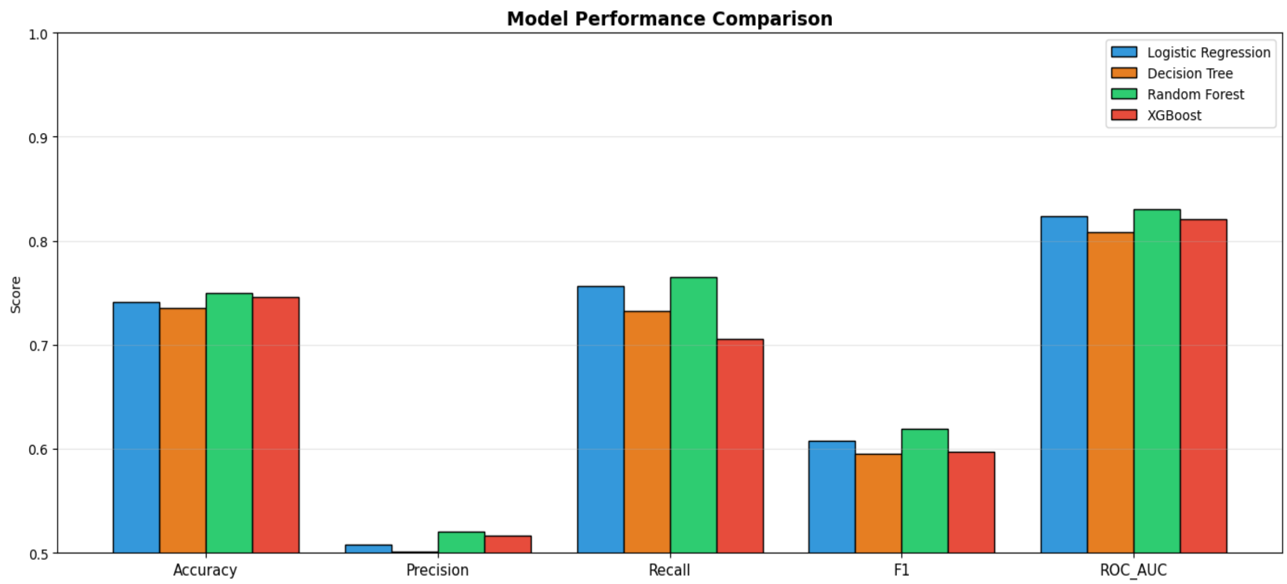

Figure 9: Comparative performance of four classifiers. Random Forest achieves the highest scores on Recall (0.765), F1 (0.619), and ROC-AUC (0.831).

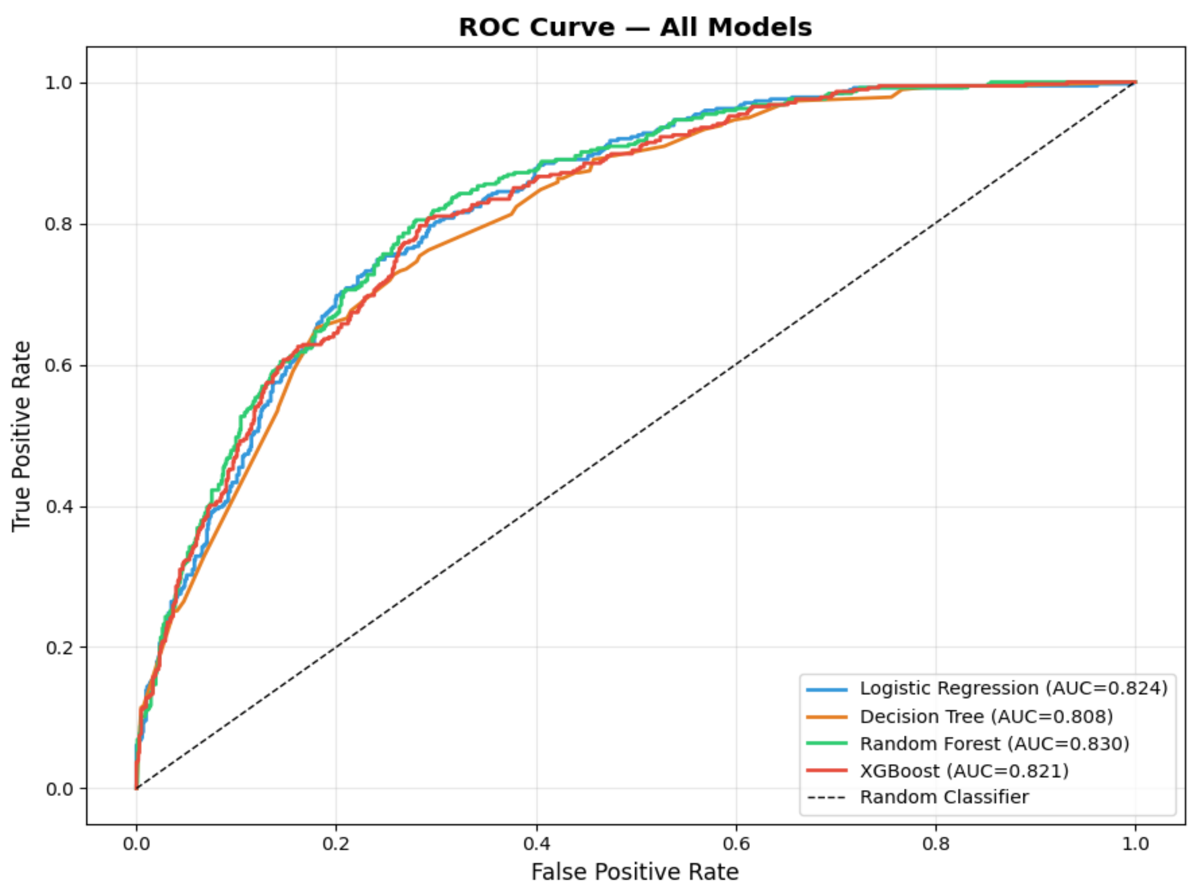

Figure 10: ROC curves for all four classifiers. Random Forest (AUC = 0.831) achieves the largest area under the curve, followed by Logistic Regression (AUC = 0.824).

Random Forest was the best in four out of five measures. The 5-fold stratified cross validation of the SMOTE re-sampled training set yielded a mean ROC-AUC of 0.847 (SD = 0.012), indicating stable generalization. The model was able to accurately predict 268 out of 374 churned customers correctly (recall = 78.1%) and missed 106 churned customers (false negatives) and 248 non-churned customers (false positives).

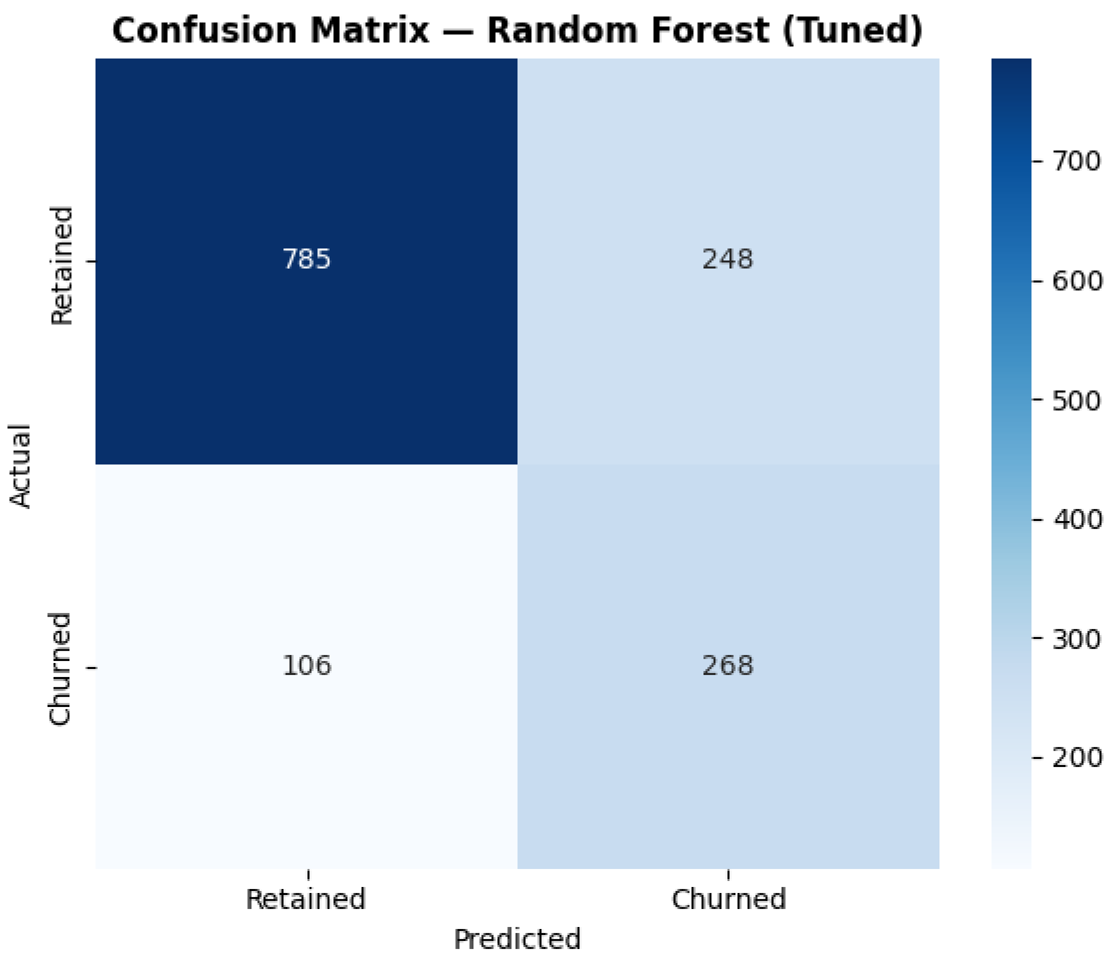

Figure 11: Confusion matrix for the Random Forest classifier at threshold 0.51. Of 374 true churners, 268 are correctly identified (recall = 78.1%).

Although XGBoost is generally considered to be better than Random Forest on tabular data, it performed worse than Random Forest in this case. This effect can be traced back to the conflict between SMOTE resampling and gradient boosting: SMOTE creates a balanced training set, which changes the residual distribution in a manner that leads to gradient boosting making corrections to the synthetic samples rather than real customer trends. In Random Forest, the bagging procedure assigns equal weights to all of the training samples, making it more resilient to the distributional changes caused by SMOTE. If you use SMOTE, try out bagging ensembles over boosting variants, otherwise XGBoost is an expected best performer.

4.3 SHAP Feature Importance

Table 3 presents the top 10 features ranked by mean absolute SHAP value, derived from the 500-instance test sample.

Rank	Feature	SHAP Score	Direction	Business Interpretation
1	IsLongTermContract	0.0721	Protective	Absence of long-term commitment is the single strongest churn driver
2	ChargesPerTenure	0.0552	Risk	High charges relative to tenure signals a perceived value deficit
3	Contract_Month-to-month	0.0527	Risk	Month-to-month customers face no exit friction
4	InternetService_Fiber optic	0.0446	Risk	Fiber subscribers experience value-expectation misalignment
5	Contract_Two year	0.0318	Protective	Two-year contracts create behavioral and cognitive commitment
6	tenure	0.0287	Protective	Established customers have accumulated switching costs
7	MonthlyCharges	0.0241	Risk	High billing increases price-sensitivity and comparison behavior
8	NumAddons	0.0198	Protective	Add-on services increase service depth and switching cost
9	OnlineSecurity	0.0176	Protective	Security add-on reduces perceived vulnerability to switching
10	TechSupport	0.0164	Protective	Technical support subscription indicates higher service engagement

Table 3: Top 10 features by mean absolute SHAP value. SHAP scores are derived from a 500-instance test sample using TreeExplainer.

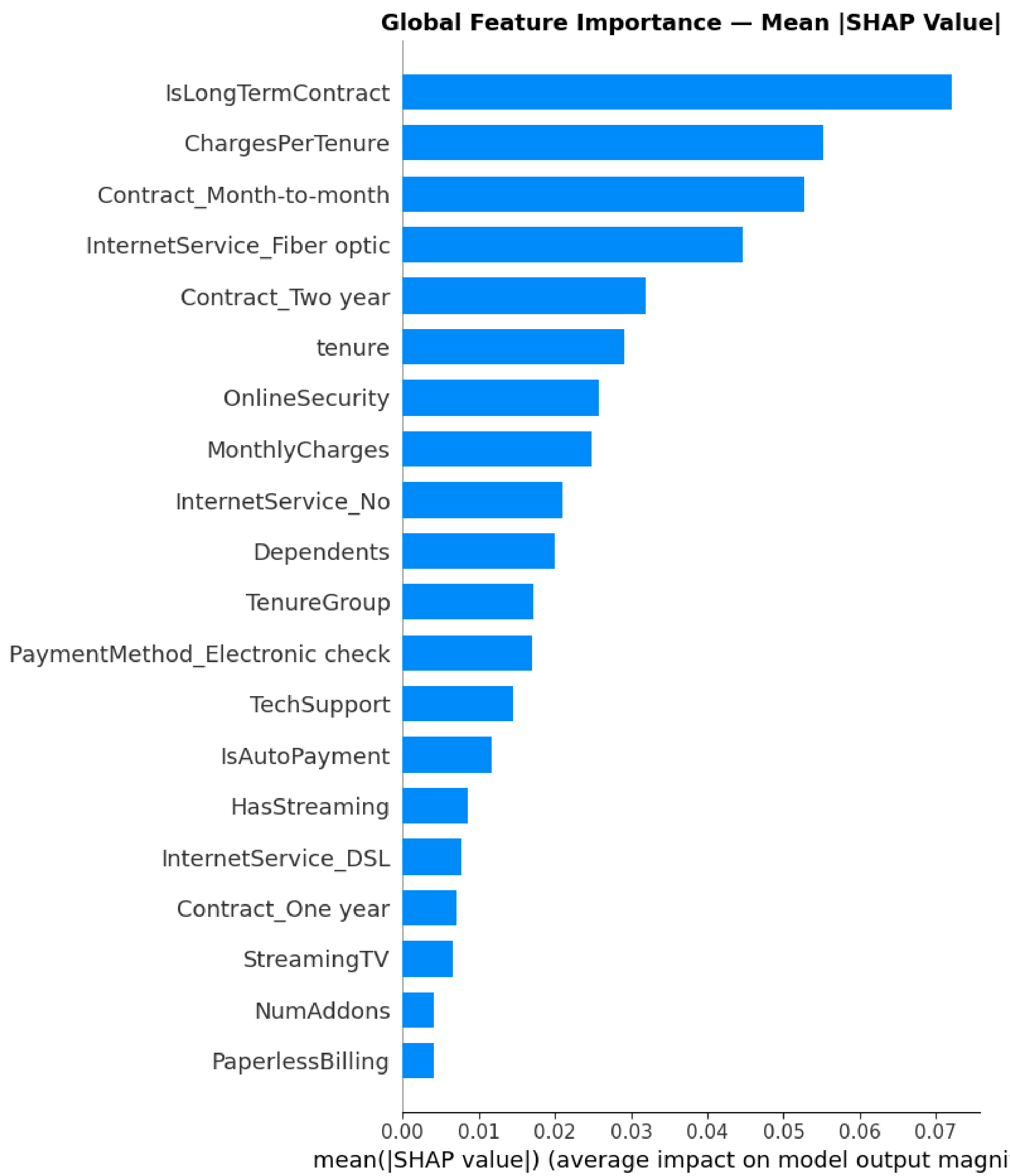

Figure 12: Global SHAP feature importance. Contract commitment and value-per-tenure ratio are the two dominant churn predictors.

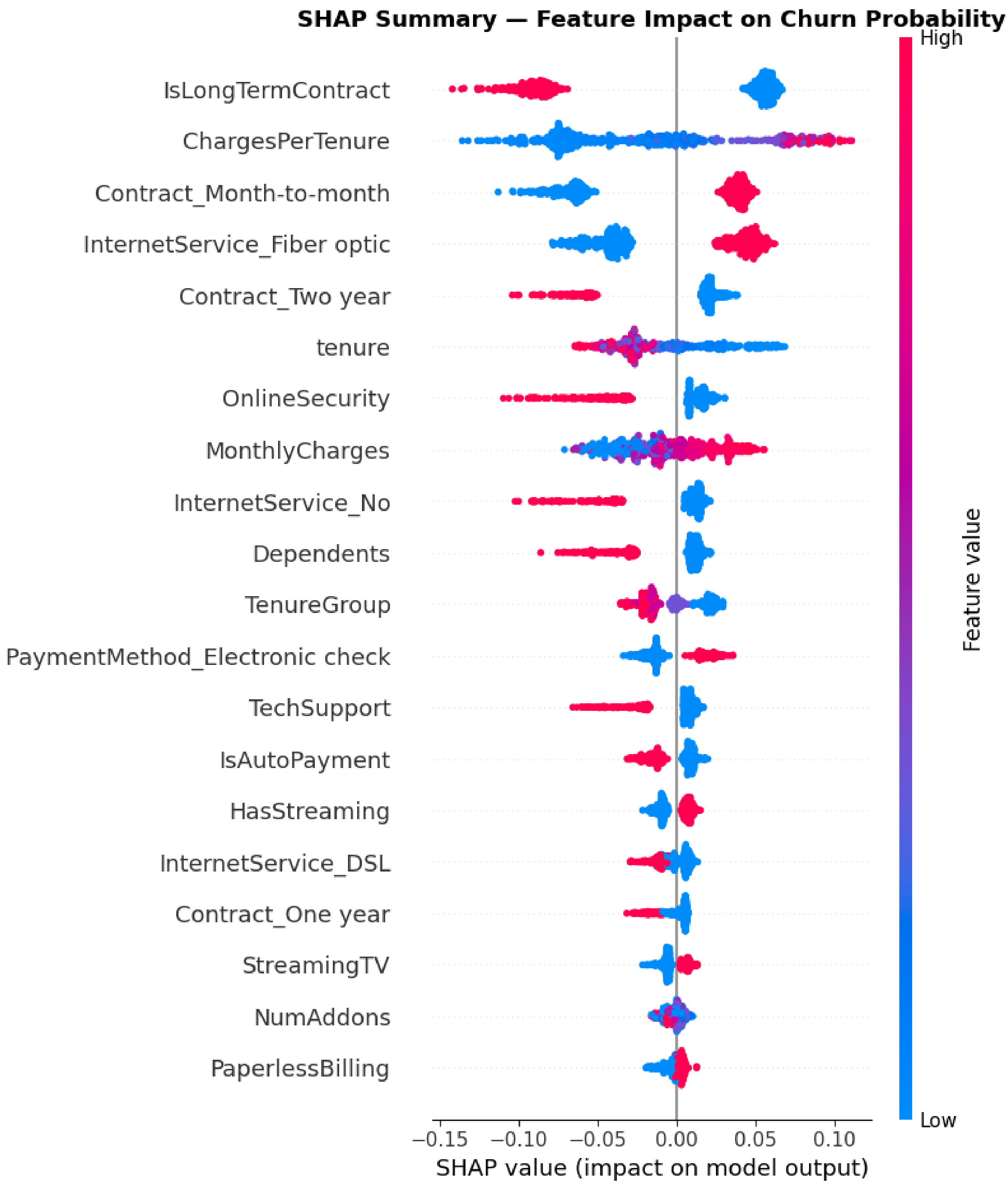

Figure 13: SHAP summary plot showing feature effect directions and distributions. High ChargesPerTenure and month-to-month contract push predictions toward churn; high tenure and long-term contract push away from it.

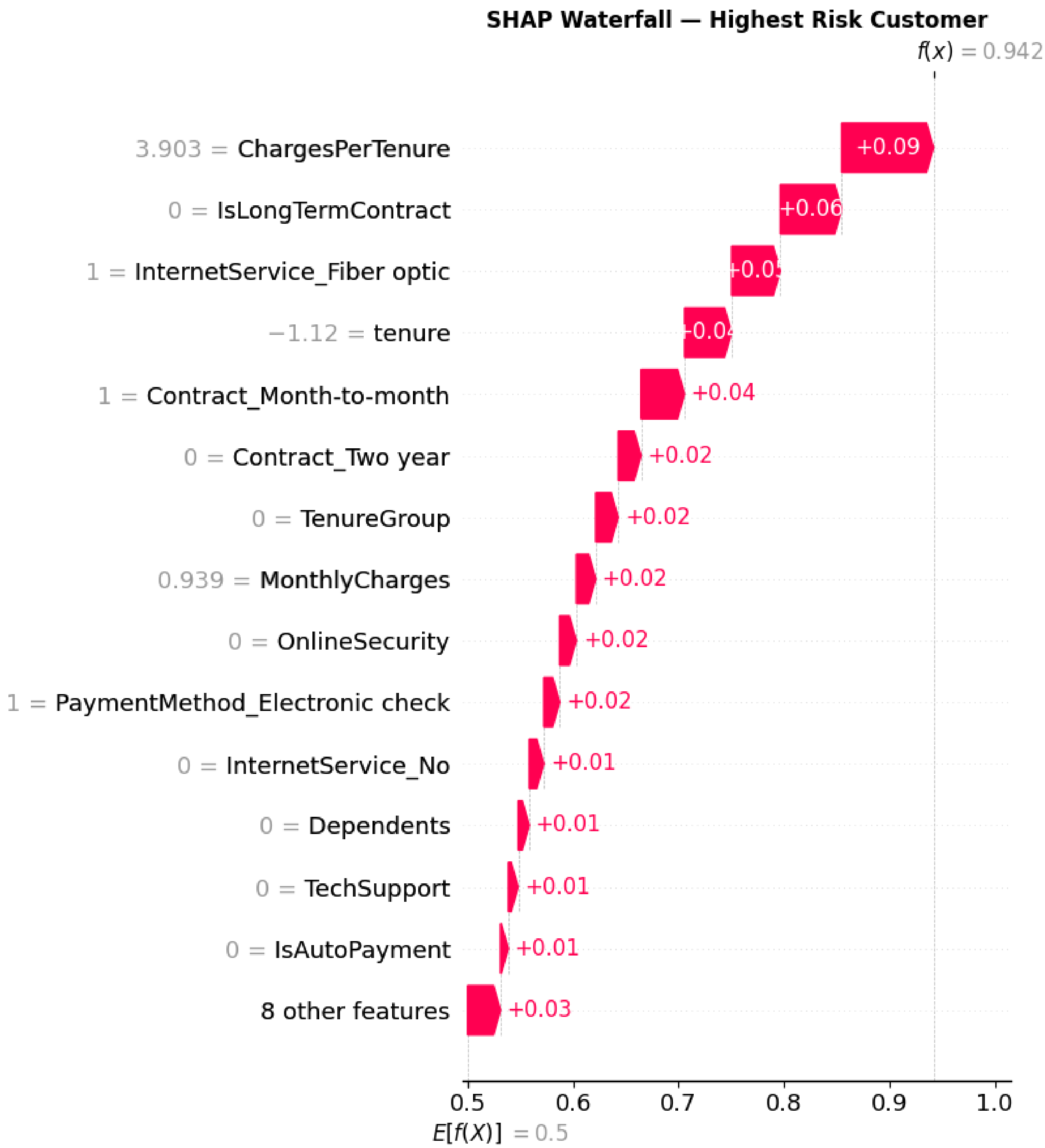

Figure 14: SHAP waterfall plot for a representative high-risk customer. Month-to-month contract and elevated ChargesPerTenure contribute the largest positive push toward churn probability.

IsLongTermContract is the top predictor with a SHAP score of 0.0721 up 31% from ChargesPerTenure at the second spot, which has a SHAP score of 0.0552. Contract_Two year is a protective driver (rank 5), while Contract_Month-to-month is a positive churn driver (rank 3) - it's the only feature that has both positive and negative SHAP values in the top five features, making it a bidirectional indicator that contract type is the most actionable lever for retention. The engineered feature ChargesPerTenure far outperforms raw MonthlyCharges (rank 7) in terms of being the most important churn driver – value perception, not bill amount alone.

4.4 Customer Segmentation Results

The Silhouette Score that was obtained for K-Means clustering, K = 4 was 0.228. The statistical optimum was K = 2 (Silhouette = 0.234) but the difference of 0.006 was accepted as worthwhile in order to gain four behaviourally distinct segments with meaningful differentiation in the retention strategy. The resulting segment profiles are shown in Table 4.

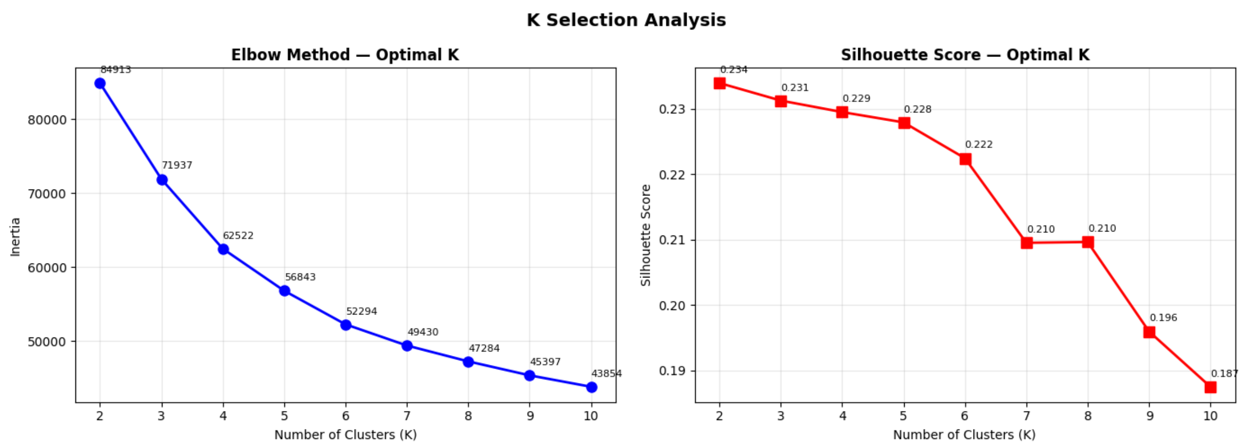

Figure 15: Silhouette coeffici Figure 15: K Selection Analysis, Elbow Method (left) and Silhouette Coefficient (right) for K = 2 to 10. The Silhouette Coefficient peaks at K = 2 (0.234) and registers 0.228 at K = 4. K = 4 was selected over the statistical optimum based on business interpretability, as K = 2 yields insufficient segment granularity for differentiated retention strategy design.ent by number of clusters (K = 2 to 10). K = 2 yields the highest silhouette score (0.234), indicating optimal cluster separation.

Table 4 presents the four customer segments produced by K-Means clustering (K = 4).

Segment	N	Churn Rate	Avg. Tenure	Monthly Charge	Add-Ons	LT Contract	Fiber %
Critical Churn-Risk	2,126	**54.7%**	21 mo	$87	1.9	1%	100%
Vulnerable High-Risk	1,744	28.5%	13 mo	$41	1.1	3%	0%

Segment	N	Churn Rate	Avg. Tenure	Monthly Charge	Add-Ons	LT Contract	Fiber %
Stable Mid-Risk	2,029	9.2%	56 mo	$84	4.1	96%	46%
Loyal Low-Risk	1,133	2.0%	42 mo	$25	0.2	100%	2%

Table 4: Customer segment profiles produced by K-Means clustering (K = 4, Silhouette = 0.187, n = 7,032).

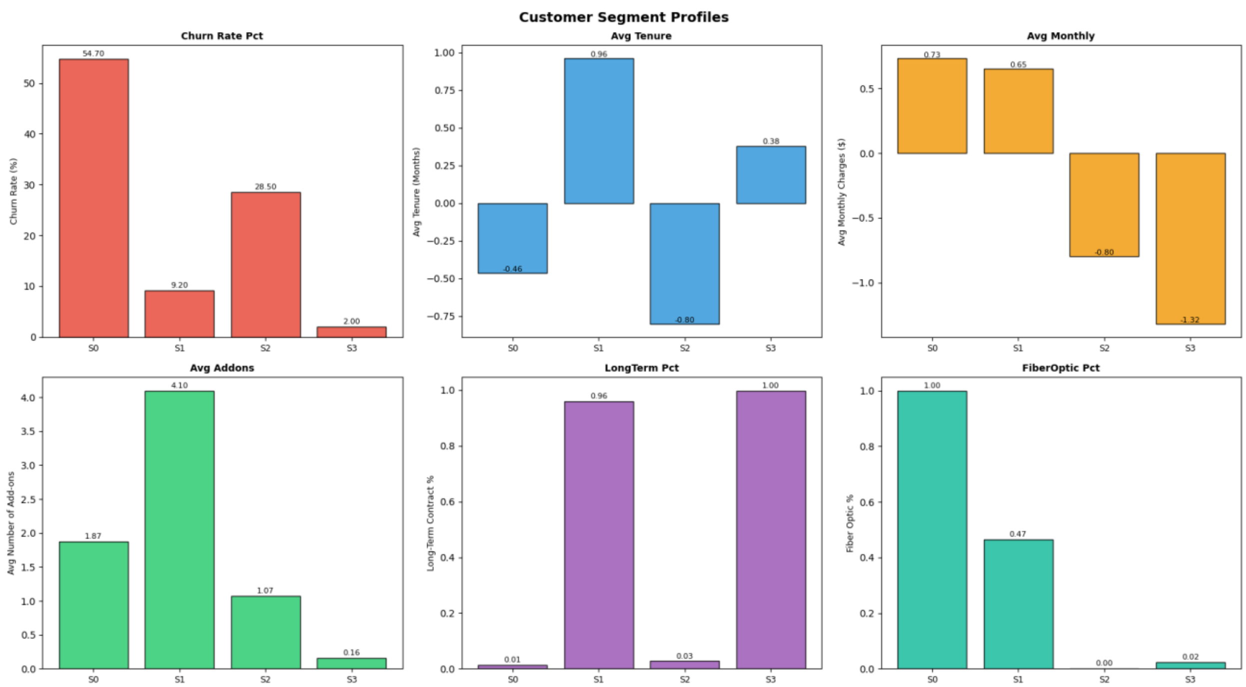

Figure 16: Segment profile comparison across key behavioral dimensions. Critical Churn-Risk customers present the worst profile across five of six dimensions.

The Critical Churn-Risk segment is characterized by its short tenure (21 months), high monthly charges ($87), low adoption of add-ons (1.9 services), low long-term contract adoption (1%) and high universal fiber optic subscription (100%). This profile fully reflects the exact match between the risk factors identified by SHAP analysis, confirming the internal consistency of the predictive and segmentation parts.

The Loyal Low-Risk segment (16.1% of customers and 2.0% churn) provides a useful contrast: although this segment has the lowest monthly charges of all segments ($25), and relatively few add-on customers (0.2), this segment has 100% long-term contract sign-up and near-zero churn.

This indicates that maintaining loyalty in this group is most likely achieved through contractual commitment, and not through service depth or price.

4.5 Retention Recommendations

Table 5 shows the priority distribution across the 7,032 processed customers.

Priority Tier	Customer Count	Share of Base	Segment	Primary Intervention Logic
Critical	2,126	30.2%	Critical Churn-Risk	Immediate outreach; contract conversion incentive; add-on trial offer
High	1,744	24.8%	Vulnerable High-Risk	Proactive check-in; auto-pay incentive; first add-on subsidy
Medium	2,029	28.9%	Stable Mid-Risk	Loyalty acknowledgment; upsell to additional add-ons
Low	1,133	16.1%	Loyal Low-Risk	Annual reward; referral program enrollment

Table 5: Retention priority distribution across the customer base (n = 7,032).

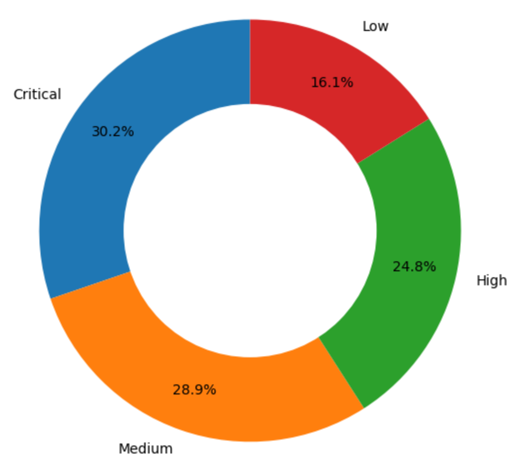

Figure 18: Retention priority distribution. Approximately 55% of the customer base (Critical + High tiers) requires active retention intervention.

Sample offers for a Critical Churn-Risk customer who has less than 12 months on the contract: call with dedicated agent to onboard immediately; monthly discount of 20% for 3 months; incentive to extend to annual contract after 2 months, for 20% discount; bundle OnlineSecurity +

TechSupport at 50% for 3 months; automatic payment with 5% discount. Annual customer reward and customer invitation to the Referral Program for Loyal Low-Risk customer.

5. KEY FINDINGS SUMMARY

Table 6 synthesizes the primary findings across all pipeline phases, connecting each insight to its supporting evidence and business implication.

Finding	Supporting Evidence	Business Implication
Contract type is the single most actionable churn driver	SHAP rank 1 (IsLongTermContract, score 0.0721); churn rate differential: 43.0% vs. 2.8%	Incentivizing contract upgrades generates the highest expected retention ROI per dollar spent
Early tenure is the highest-risk lifecycle stage	Churn rate 47.4% in months 0–12; Critical segment avg. tenure = 21 months	Dedicated onboarding programs and introductory discounts should be standard practice for new subscribers
Value perception drives churn more than billing amount	ChargesPerTenure ranks 2nd in SHAP (0.0552), above raw MonthlyCharges (rank 7, 0.0241)	High-charge new customers are a distinct risk cohort requiring early value demonstration
Fiber optic service carries disproportionate churn risk	41.9% churn rate vs. 18.9% for DSL; fiber subscribers make up 100% of Critical segment	A structural value-delivery gap exists for fiber customers; requires targeted quality and value interventions
Add-on adoption functions as a retention anchor	SHAP rank 8 (NumAddons, 0.0198); churn rate declines from 31.7% (0 add-ons) to 7.8% (4+ add-ons)	Early add-on adoption should be integrated into onboarding to accelerate switching cost accumulation
SMOTE favors Random Forest over XGBoost	RF ROC-AUC 0.831 vs. XGBoost 0.821; distributional shift from SMOTE conflicts with gradient boosting residual weighting	Practitioners using SMOTE should benchmark bagging ensembles alongside boosting methods
55% of the customer base requires active retention intervention	Critical (30.2%) + High (24.8%) segments combined; average 4.2 recommendations per Critical customer	Retention resource allocation should be concentrated on these two segments, not distributed uniformly

Table 6: Key findings summary across all pipeline phases, with supporting evidence and business implications.

6. DISCUSSION

6.1 Predictive Model Findings

What is counterintuitive here is that this dataset is a structured tabular one and Random Forest generally outperforms XGBoost when it comes to structured tabular data, which needs to be explained. Gradient boosting fits the residuals sequentially; each tree focuses on the errors made by the ensemble of trees built so far, with the scale of the focus depending on the distribution of the residuals. If SMOTE creates a balanced training set by synthesizing minority-class samples, then it will modify this remaining distribution in a manner that is not representative of the data generating process. These amplifications to "false" synthetic samples by the boosting algorithm inevitably lead to a deterioration of generalization to the original-distribution test set.

Random Forest's bagging technique assigns equal weights to all the training samples, which makes it structurally stronger against the distributional shift that is brought by SMOTE. Given the acceptable, structurally predictable overfitting (train-test AUC gap of 0.016), the current study provides supporting evidence for this error.Current study results support this error, which are acceptable and structurally predictable (train-test AUC gap of 0.016), with the training AUC of 0.847 and test AUC of 0.831. SMOTE adds synthetic samples which are easier to be classified as the class compared to the held-out sample, artificially raising training AUC. The operationally relevant test AUC is 0.831.

6.2 Explainability Findings

The engineered ChargesPerTenure feature is the second most important feature in terms of SHAP score, behind raw MonthlyCharges, with a score of 2.3× of MonthlyCharges, confirming the feature engineering hypothesis. Churn isn't just a function of bill, it's a function of bill vs. tenure. A customer who pays $80 per month at the end of 6 months will have a different value perception than one who pays the same at the end of 5 years: the first has not yet received enough value to justify the amount that he or she is paying, the latter has incurred switching costs and familiarity with the service. The finding has obvious product design implications: High dollar new subscribers are a specific risk group that must be shown the value, not just reduced billings.

The bidirectional contract type evidence, that month-to-month is a positive churn driver (rank 3) and the two-year contract is a protective driver (rank 5) indicates that this is not just correlational, but actually a commitment mechanism. With month-to-month customers, you are completely flexible to leave without any exit fee, and with two-year contract customers, you will have the contractual enforcers and the signalled prior commitment that helps to reinforce the customer's decision to stay. This has implications for Product Design: Friction on the month to month trade is a low ROI design intervention, as is increasing the value difference between contract tiers.

6.3 Segmentation Findings

The Critical Churn-Risk segment of the data set is an apparent contradiction: the customers in this segment pay the highest amount of money per month ($87) and churn the most (54.7%). This fits into a value perception failure model: customers have been "sold" high-end fiber optic service at a high price point but have not yet achieved enough tenure (21 months) to feel part of a service ecosystem. No add-on switching costs (average 1.9 services) and neither any month-to-month contracts nor any exit friction. For this segment, it is crucial to retain them while also increasing the perceived value of adding-on and driving the need to upgrade contracts.

The Loyal Low-Risk segment is characterized by the low level of add-on adoption (0.2), low charges ($25) and close to zero churn (2.0%) as contract loyalty is often enough to secure loyalty without strong service engagement. This means that in some instances, retention investment is not about providing service enrichment, but rather about contract reinforcement for loyal customers and recognition of their tenure.

6.4 Limitations

There are three limitations of this study. First, it is a cross-sectional dataset, which just collects customer characteristics at one moment in time without looking at their behavior over time. There are no dynamic churn signals, no falling usage, no rising support contact, no payment irregularities; if they were added to a longitudinal model it would probably result in a significantly higher predictive accuracy.

Second, the recommendation engine is a rule-based instead of a learned one. The system can't optimize based on observed outcomes if there's no feedback data to guide them on which interventions convert into which customers. Third, the data is from one hypothetical company and not applies to the competitive environment, regional factors, or macroeconomic considerations that are relevant in real-world telecom markets. The external validity must be validated on actual operators' datasets.

7. RECOMMENDATIONS FOR TELECOM OPERATORS

7.1 Prioritize Contract Conversion

Customer churn is 43.0% for month-to-month contract customers, compared to 2.8% for the two-year contract customers, with 55.1% of customers on month-to-month contracts. The most important feature in SHAP is contract type, which scores 0.0721. Conclusions: SHAP analysis (Table 3, ranks 1, 3, 5); EDA churn rate differential (Section 3.4).

Recommendation: Implement a graduated discount on contract length upgrades. Use pro-active outreach to month-to-month customers as they approach the 6-month and 12-month tenure landmarks, when a customer is likely to face a 47.4% churn risk. Given the 31.8-point churn rate difference between one-year and month-to-month contracts, a 10% conversion rate of month-to-month customers to one-year contracts in the Critical segment would drop that segment's churn rate by an estimated 4–6 percent.

7.2 Establish Early-Tenure Intervention Programs

Issue: 47.4% of customers in the cohort of 0–12 months churn while only 6.1% churn after four years of tenure. The average tenure of the Critical Churn-Risk segment is 21 months.

Evidence: consists of the following: Tenure group churn gradient (Figure 7, Section 4.1); Critical segment profile (Table 4).

Recommendation: Create a new subscriber onboarding program which includes a new subscriber agent, a satisfaction check-in at month one, an offer for the first three months, and a proactive add-on trial offer at month two. Focus on the highest risk combination (fiber optic + month-to-month + electronic check) for higher intensity of interventions. Highest leverage intervention is expected to be reducing early tenure churn. The estimated 47.4% to 35% reduction in the 0-12 month cohort churn rate would result in the recovery of about 130 customers per 1,000 new subscribers

.7.3 Address Fiber Optic Value Delivery

Issue: Fiber subscribers are churning at 41.9% while paying premium charges. Fiber subscription is ranked as the 4th SHAP importance (value = 0.0446). The Critical Churn-Risk segment has 100% fiber penetration.

Evidence: EDA findings (Section 3.4); SHAP Table 3 rank 4; Critical segment profile (Table 4).

Recommendation: Adopt proactive service quality monitoring with automatic outreach triggers for fiber customers that are having any service degradation. Targeted value demonstrations (speed comparisons, service reliability reporting) of new fiber subscribers (months 1-6). To lower the ChargesPerTenure in the crucial opening phase, you might want to consider using a fiber-specific introductory pricing period..

7.4 Accelerate Add-On Adoption During Onboarding

Issue: The churn rate is 31.7% for customers with no add-ons versus 7.8% for customers with four or more add-ons. The Critical Churn-Risk segment has an average of just 1.9 add-ons. The importance of NumAddons is 0.0198 (Rank = 8).

Evidence: EDA add-on churn gradient, Figure 8: Section 4.1; SHAP Table 3 rank 8; Critical segment profile (Table 4).

Recommendation: Offer a bundle of OnlineSecurity and TechSupport for the first 3 months at 50% off for new subscribers. Consciously set up the trial as service enrichment, not upsell. Automate a reminder to renew the discounted subscription at 90-days, when switching costs start to add up and become measurable value for retention. Based on the observed churn gradient, an expected impact of moving a Critical segment customer from 1.9 to 3+ add-ons is a reduction of about 15-20 percentage points in segment level churn rate.

8. CONCLUSIONS

8.1 Summary of Findings

This study developed and validated an end-to-end telecom customer churn prediction and retention recommendation system integrating supervised machine learning, SHAP explainability, K-Means customer segmentation, and a prescriptive recommendation engine within a single deployed pipeline. Key outcomes are as follows.

- Random Forest achieved the optimal balance of recall (78.1%) and ROC-AUC (0.831) across four classifiers under SMOTE-resampled conditions, outperforming XGBoost due to the compatibility of its bagging mechanism with SMOTE-altered training distributions.
- Contract type is the primary churn driver in both directions: absence of long-term contract is the strongest positive churn predictor (SHAP rank 1), while two-year contract adoption is the strongest protective predictor among the top five.
- The engineered ChargesPerTenure feature ranked second in global SHAP importance, above raw MonthlyCharges, confirming that value perception, charges relative to tenure, is a more operative churn mechanism than billing amount in isolation.
- K-Means clustering produced four coherent segments with churn rates from 2.0% to 54.7%, aligned with the SHAP-identified risk factors, validating pipeline internal consistency.
- Approximately 55% of the customer base requires active retention intervention (Critical + High tiers), concentrated in the first two years of tenure and among fiber optic subscribers without long-term contracts.

8.2 Broader Implications

Three broader findings extend beyond this specific dataset. First, SMOTE-resampled training conditions systematically favor bagging ensembles over gradient boosting; this should be treated as an empirical benchmark decision rather than an assumption. Second, domain-informed feature engineering can produce predictors that outperform original dataset variables in SHAP importance, justifying the engineering investment even on well-curated benchmarks. Third, integrating prediction, explanation, segmentation, and prescription within a single deployed system is both technically feasible at intermediate complexity and substantially more operationally useful than any single component in isolation.

8.3 Future Research Directions

Four directions are identified for future development:

- Longitudinal churn modeling incorporating time-series behavioral signals (usage patterns, support contact frequency, payment irregularities) using recurrent neural networks or temporal convolutional architectures.
- Reinforcement learning optimization of the recommendation engine, enabling the system to update intervention rules based on observed retention outcomes.
- Causal inference methods (propensity score matching, instrumental variable analysis) to estimate the counterfactual retention impact of specific interventions, moving beyond SHAP correlation to causal attribution.
- Multi-operator validation using datasets from actual telecommunications companies across diverse markets, to establish the external validity of the identified churn drivers and model selection findings.

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